

Chapter 129 — Tumours of the Urinary Bladder

EPIDEMIOLOGY

Incidence, Prevalence, and Mortality

- **A global disease:** about 550,000 new cases and 200,000 deaths worldwide in 2018; in the United States (US) about 82,000 diagnoses and 17,000 deaths a year.
- **Age-adjusted incidence and mortality fell** through the 1990s and early 2000s and have since plateaued.
- **Crude incidence is rising** because of population ageing and growth; adjusted for demographic change, incidence is stable, and the US and Europe show a slow decline.
- **Risk is highest in high-income countries** and lowest in low-income countries; the highest regional rates are in North America and Western and Central Europe, the lowest in Oceania and Andean and Central Latin America.
- **The burden is shifting to the developing world:** more than 60% of incident cases and nearly half of deaths occur there, attributed to a narrowing gap in life expectancy between low- and high-income countries.
- **Disability-adjusted life years (DALYs)** are the World Health Organization (WHO) measure of disease burden: years of life lost (deaths $\times$ remaining life expectancy) plus years lived with disability. Bladder cancer caused about 3.4 million DALYs in 2015; age-standardised DALY rates fell in both sexes.
- **Prevalent out of proportion to incidence:** deaths are few relative to cases, so bladder cancer is the **fourth most prevalent cancer** affecting both sexes globally and in the top five in countries of every income level. Five-year prevalence understates the number of survivors still managing disease sequelae.
- **Regional variation** reflects demography and environmental exposure: bladder cancer was the **commonest cancer in Egyptian males** (2015) because of *Schistosoma haematobium*, which causes squamous cell carcinoma (SCC); in Europe the highest rates are in **Belgium (males) and Hungary (females)**.

Fig. 129.1 — Global incidence (age-standardised rate, ASR), 2018, by sex (GLOBOCAN; Ferlay et al., 2014).

- **Highest ASRs** in North America, Europe and Australia for both sexes; female rates are several-fold lower in absolute terms, with the same geographic pattern.

Fig. 129.2 — Trends in age-standardised incidence, 1990–2015, by sociodemographic index (SDI) (Fitzmaurice et al., 2017).

- **Rates rise with SDI:** high-SDI countries have the highest and slowly declining incidence; male rates are far above female rates at every SDI level.

Fig. 129.3 — Age-specific incidence, United Kingdom (Cancer Research UK).

- **Rates climb steeply from the fifth decade** and are highest at age 90 and over; absolute case numbers peak in the eighth and ninth decades.

Gender, Racial, and Age Differences

- **A disease of ageing and cumulative exposure:** average age at diagnosis 73 (US); about 9 in 10 patients are diagnosed after 55. Incidence rises from age 50–54 in both sexes, more sharply in males from 60–64, and is highest at 90 and over.

- **Male predominance about 3:1:** nearly three-quarters of cases are in males. US lifetime risk is 1 in 27 for males and 1 in 89 for females.
- **Why males — smoking:** smoking is far more prevalent in males worldwide.
- **Why males — carcinogen metabolism:** the disparity **persists after controlling for smoking**. Uridine 5'-diphospho-glucuronosyltransferase (UGT; aromatic amine metabolism) and glutathione S-transferase mu 1 (GSTM1; detoxification) are expressed differently in the sexes, so equal pack-years may yield unequal carcinogen exposure.
- **Why males — sex steroids:** in females, **later menarche (≥15 years), parity and oestrogen or progestin therapy** are associated with lower risk. The androgen receptor (AR) is downregulated in bladder tumours, more so at higher stage and grade, and **castration or AR knockout suppresses tumour growth** in transgenic and carcinogen-induced mouse models.
- **Females present later and do worse stage for stage:** they lose a larger share of remaining life expectancy, and stage at presentation explains only about 30% of the disparity.
- **Causes of the female disadvantage:** tumour biology, environmental exposure and **implicit bias** (implicit social cognition — bladder cancer framed as a "male" disease), with a **longer delay from first haematuria to diagnosis** because females are more often managed as urinary tract infection (UTI), which delays referral to a urologist.
- **Race:** incidence is highest in White Americans (1.5 times that of Black, twice that of Hispanic and six times that of Native Americans), but **Black patients more often present with muscle-invasive disease**, more often have **nonurothelial histology (highest in Black females)**, and have worse stage-for-stage 5-year relative survival.

MALES VERSUS FEMALES — THE GOVERNING ASYMMETRY

	Males	Females
Incidence	About three times higher	Lower
Stage at presentation	Less advanced	More advanced
Stage-for-stage outcome	Better	Worse
Diagnostic pathway	Haematuria investigated promptly	Haematuria attributed to UTI; delayed referral
Protective factors	—	Later menarche, parity, exogenous oestrogen or progestin

Fig. 129.4 — Five-year relative survival by race and stage, US 2007–2013 (Siegel et al., 2018).

- **Survival falls steeply** from in situ and localised to regional to distant disease, and **Black patients fare worse at every stage**.

Economic Impact

- **The most expensive cancer to treat per patient over a lifetime:** lifetime cost roughly US\$129,000–251,000 per patient; annual direct costs exceed US\$4 billion in the US and €4.9 billion in the European Union.
- **Costs vary several-fold between countries** and between institutions, driven mainly by **differences in the management of non-muscle-invasive bladder cancer (NMIBC)**; per capita, the US spends about twice as much as Europe.

- **Two targets for cost reduction:** intensive NMIBC surveillance (annual procedures and laboratory tests) and metastatic disease (costly late-stage therapeutics and end-of-life care).

Concept — *Prevalence and cost follow from the same fact: bladder cancer is common, rarely fatal early and repeatedly recurrent, so patients accumulate years of surveillance; the rising burden reflects an ageing population, not a rising age-specific risk.*

RISK FACTORS

- **An exposure surface:** like skin and lung, the bladder is in constant contact with environmental carcinogens and inflammation.
- **Aromatic amines** (2-naphthylamine, 4-aminobiphenyl, benzidine) are the **central causative agents**; once hydroxylated they form DNA adducts. Tobacco and occupation are the two commonest routes of exposure.

Smoking

- **The main known cause:** tobacco accounts for **30–40% of urothelial carcinoma**; smokers carry a **twofold to threefold risk**. There are more than 1 billion smokers worldwide.
- **Cigarettes are commonest**, but pipe and cigar smoking also raise risk; bladder carcinogens have been found in **electronic cigarette users**, suggesting possible risk. US adult smoking prevalence fell substantially between 2005 and 2015.
- **N-acetyltransferase 2 (NAT2) slow acetylators** are at particularly high risk when they smoke, because NAT2 acetylation detoxifies aromatic amines.
- **Supplements do not protect:** dietary beta-carotene may be protective, but supplemental beta-carotene and vitamins C, E or folate do not reduce risk in exposed people.
- **Cessation lowers risk progressively:** risk remains raised 1–3 years after stopping and approaches that of never-smokers after more than 15 years.
- **About 20% are still smoking at diagnosis**; active smoking raises **local recurrence in NMIBC** and **cancer death in muscle-invasive bladder cancer (MIBC)**, and many continue after diagnosis.
- **Quitting after diagnosis:** its survival benefit is unproven, but continued smoking raises the risk of a second primary cancer, peripheral vascular disease and chronic obstructive pulmonary disease (COPD).

Occupational Risk

- **Occupation accounts for 5–10% of bladder cancer**; lifetime risk is greatest in work with chemicals and dyes.
- **Exposure among females is rising** as many of these occupations become increasingly female.

OCCUPATIONAL CARCINOGEN CLASSES

Carcinogen class	At-risk occupations
Aromatic amines	Tobacco, dye and rubber workers; hairdressers; painters; leather workers
Polycyclic aromatic hydrocarbons	Chimney sweeps, nurses, waiters, aluminium workers, petroleum workers, seamen

Fig. 129.5 — Summary relative risks of bladder cancer by occupation (meta-analysis) (Cumberbatch et al., 2015).

- **Highest relative risks:** tobacco workers, dye workers, chimney sweeps, nurses, rubber workers, waiters and aluminium workers. **Lowest** (below 1): farmers, gardeners, teachers and forestry workers. Overall excess is modest and heterogeneity very high.

Environment

- **Arsenic in drinking water** (a toxic metalloid released from weathering rock) is associated with bladder cancer. The worst-affected regions are **West Bengal, Bangladesh and Taiwan** (and parts of Chile), with concentrations hundreds of times the **recommended limit of 10 µg/L**.
- **Mechanism is multifactorial**: oxidative stress, epigenetic effects and impaired DNA repair.

Radiation

- **First recognised in atomic bomb survivors**, in whom attributable risk rose with dose.
- **Latency 15–30 years**; radiation-associated bladder cancer is not age related.
- Studies are conflicting and confounded, but **risk scales with modality**:

RADIATION MODALITY AND SECONDARY BLADDER CANCER

Modality	Relative risk
External beam radiation therapy (EBRT), cervical cancer	Twofold to fourfold
EBRT, prostate cancer	1.5- to 2.0-fold
Brachytherapy	Smaller
Intensity-modulated radiation therapy (IMRT)	Little or none

Genetics

- **NAT2 and GSTM1 deletion** are the most-studied genes; both govern aromatic amine metabolism and so matter chiefly in those with carcinogen exposure. NAT2 regulates acetylation of the aromatic amines in cigarette smoke.
- **Several low-penetrance variants** probably combine with occupational, environmental and behavioural exposures to potentiate risk.

Hereditary

- **First-degree relatives have a twofold risk**, but high-risk families are exceedingly rare; there is **no mendelian pattern**, so linkage studies are unhelpful — the aetiology is gene–environment interaction.
- **Lynch syndrome** (hereditary nonpolyposis colorectal cancer, HNPCC) raises urothelial cancer risk, **mainly in MSH2 carriers**.
- **Pathogenic germline variants in 13–24%** of patients with urothelial carcinoma (mostly advanced disease), commonest **MSH2 and BRCA1/2**. In NMIBC they occur in high-grade but **not low-grade** disease, so **germline testing may help in selected high-grade patients** to identify risk in family members.

Body Mass Index

- **Increasing body mass index (BMI) raises risk** in multiple meta-analyses, with a **dose–response** relationship, and **independent of smoking**.
- **Obesity worsens prognosis** after diagnosis: more recurrence and a shorter time to recurrence.
- **Proposed mechanisms**: insulin resistance, chronic hyperinsulinaemia, increased bioavailability of steroid hormones and local inflammation with proinflammatory cytokines; the exact mechanism is unknown.

Medical Conditions

- **Chronic inflammation → keratinising squamous metaplasia → SCC** is the usual direct pathway.

- **Neurogenic bladder and spinal cord injury with chronic indwelling catheters** carry a modestly increased SCC risk; bladder calculi, outflow obstruction, recurrent UTI and catheter trauma are thought additive.
- **Bladder exstrophy**: malignancy in **3.3–7.5%**, several times age-matched controls; **adenocarcinoma accounts for more than 90%**, SCC and urothelial carcinoma the remainder.
- As patients with spinal cord injury and congenital bladder anomalies live longer, **bladder cancer rates in these groups may rise**.

Schistosomiasis

- **Endemic in up to 76 developing countries**, affecting up to 218 million people, and a major cause of bladder cancer in tropical countries despite WHO eradication campaigns.
- **Life cycle**: blood flukes with mammalian hosts and freshwater snails as intermediate hosts. Of the four human schistosomes, **only *S. haematobium* is linked to bladder SCC**.
- **Mechanism**: worms live in the **vesical venules** and lay eggs → irritation and fibrosis via a **T-helper 2 (TH2) proinflammatory response**; infection also **potentiates other risk factors, notably smoking**.

Recurrent Infection

- **Chronic UTI: a modest, confounded association** — confounded by catheterisation, stones, smoking and occupation; there are no prospective studies of UTI, and early data have not been validated in contemporary cohorts.
- **Gonorrhoea**: a prospective study found **nearly twofold risk in males**, with more invasive tumours; the mechanism may be nitrosamine production.
- **Urinary microbiome**: some bacterial and viral colonisation of the urothelium may be **protective**; mechanisms are under investigation.

EXPOSURE PATHWAY PREDICTS HISTOLOGY

Histology	Associated exposures
Urothelial carcinoma	Tobacco, aromatic amines, arsenic, radiation, cyclophosphamide, Lynch syndrome
Squamous cell carcinoma	<i>S. haematobium</i> , chronic indwelling catheter, stones, recurrent infection
Adenocarcinoma	Bladder exstrophy, schistosomiasis, chronic irritation or obstruction

Concept — Almost every risk factor is either a **chemical load** (aromatic amines, modified by acetylator and glutathione *S*-transferase genotype) or **chronic inflammation**, and the pathway predicts the histology: chemical injury gives urothelial carcinoma, chronic inflammation gives squamous or glandular tumours.

DRUGS

Pioglitazone

- **Thiazolidinedione** for non-insulin-dependent diabetes. The **PROactive randomised trial (2005)** unexpectedly found more bladder cancers with pioglitazone than placebo; later studies have both confirmed and refuted this.

Chemotherapy

- **Cyclophosphamide is the only chemotherapeutic agent shown to cause bladder cancer**; the mutagenic metabolite is **phosphoramidate mustard**.
- **Risk about 4.5-fold, dose dependent**: most pronounced above **20 g cumulative**; above **50 g**, about 7 excess cases per 100 patients treated.

- **Latency:** risk rises after 5 years and **peaks at years 10–14**.

DIET

DIETARY EXPOSURES AND BLADDER CANCER RISK

Exposure	Finding	Note
Total fluid intake	No association	Early retrospective protective signal not reproduced in the European Prospective Investigation into Cancer and Nutrition (EPIC)
Coffee and tea (caffeine)	No association	Earlier modest link was confounded by smoking
Alcohol	No association at any amount	Unlike gastrointestinal and hepatobiliary cancers
Red meat	No association (EPIC)	Unlike other malignancies
Fruit and vegetables	Lower risk in never- and former smokers, not current smokers (EPIC)	No association in meta-analyses adjusted for smoking
Vitamin and mineral supplements	Neither carcinogenic nor protective	VITamins cohort; Selenium and Vitamin E Cancer Prevention Trial (SELECT) — no effect of selenium or vitamin E

- **No specific food or food group** has been shown to mitigate risk, although many ingested carcinogens are excreted in urine and contact the urothelium directly.

Concept — *Every apparent dietary signal has dissolved once smoking is controlled for; diet is a confounder-dominated field, and no dietary intervention reduces bladder cancer risk.*

KEY POINTS

- Bladder cancer is related to **age and environmental carcinogen exposure**.
- Smoking accounts for **30–40%** of urothelial carcinoma.
- Bladder cancer is **about three times as common in males**, but **females present with more advanced tumours and a worse prognosis**.
- It is the **most expensive cancer to treat per patient** over a lifetime.
- **Pathogenic germline variants** are present in a select group of patients.
- **No dietary intervention** has reduced bladder cancer risk.

DETECTION OF UROTHELIAL CARCINOMA

- **Rarely an incidental autopsy finding**, unlike many cancers.
- **Painless gross haematuria in 85%** of newly diagnosed patients; **microscopic haematuria in nearly all**. Haematuria is typically intermittent and may follow Valsalva manoeuvres.
- **Any episode of gross haematuria must be evaluated**, even if subsequent urinalysis is negative; **antiplatelet or anticoagulant therapy does not preclude full evaluation**.
- **Irritative voiding symptoms** (frequency, urgency) may signal bladder cancer, particularly **carcinoma in situ (CIS)**.
- **After a complete negative evaluation**, the risk of malignancy with recurrent gross or microscopic haematuria is nearly zero for 6 years; for persistent haematuria, **repeat evaluation at 3–5 years** is reasonable.

- **Prostate-specific antigen (PSA)** may be considered after discussion, because about 10% of patients with recurrent gross haematuria have prostate cancer.
- **Bimanual examination** rarely finds a bladder tumour at presentation but should be performed at **transurethral resection of bladder tumour (TURBT)** if the tumour appears invasive; the history covers risk factors and family history.

GROSS HAEMATURIA VERSUS MICROHAEMATURIA — EVALUATION

	Gross haematuria	Asymptomatic microhaematuria (AMH)
Identifiable cause / malignancy risk	Cause in about 50%; malignancy about 20%	Malignancy 2.6–5.2%
Cystoscopy	All adults	All patients aged 35 years and over
Upper tract imaging	Multiphase computed tomography (CT) with delayed phase — CT urogram (CTU) — if tolerated	2012 American Urological Association (AUA) guideline: CTU; renal and bladder ultrasound may safely replace it, as upper tract malignancy is exceedingly rare
Cytology / urine markers	Reasonable	Avoid in routine evaluation (AUA)
Also	Focused history and examination; urine culture	

URINE DIPSTICK VERSUS MICROSCOPIC URINALYSIS

	Dipstick	Microscopic urinalysis
Principle	Reagent detects peroxidase activity of haem	Direct examination of sediment for red blood cells
Performance	Sensitivity 91–100%; specificity about 65% → unnecessary evaluations	Confirmatory standard
Role	Dipstick alone is discouraged for microhaematuria; confirm with microscopy	

Cystoscopy

- **Gold standard for diagnosis: cystoscopy and biopsy.** Flexible office cystoscopy is as reliable as rigid.
- **White light cystoscopy (WLC)** is the standard of care, allowing mapping and resection of all visible tumours; excellent for large papillary tumours, **less reliable for small papillary tumours and CIS.**
- **Hexaminolevulinate blue light cystoscopy (HAL-BLC):** a meta-analysis found BLC detected more Ta tumours and substantially more CIS than WLC; in about a quarter of patients CIS was seen only on BLC, with benefit in both primary and recurrent disease. **Flexible BLC in the office** finds recurrences missed on WLC in about a fifth of patients.
- **Narrow-band imaging (NBI)** improves CIS detection in mostly single-centre or underpowered studies. In a **multicentre randomised trial** of NBI-assisted versus white light TURBT, overall 12-month recurrence was similar, but **NBI reduced recurrence in low-risk patients.**

CYSTOSCOPIC IMAGING MODALITIES

	White light (WLC)	Blue light (HAL-BLC)	Narrow-band imaging (NBI)
Principle	Standard illumination	Photoactive porphyrins accumulate in neoplastic tissue → red fluorescence under blue light, 360–450 nm	Optical, no dye: blue (415 nm) and green (540 nm) bands absorbed by haemoglobin → vessels dark brown or green on pink or white mucosa

	White light (WLC)	Blue light (HAL-BLC)	Narrow-band imaging (NBI)
Agent	None	HAL; 5-aminolevulinic acid (5-ALA) studied, but only HAL is Food and Drug Administration (FDA) approved — surveillance, rigid and flexible	None
Strength	Large papillary tumours	CIS and small Ta tumours	CIS; recurrence in low-risk patients
Status	Standard of care	Adjunct	Adjunct

- **Depth of light penetration increases with wavelength** — the physical basis of NBI's selective contrast.

Urine Cytology

- **Papanicolaou (1945)** first proposed that exfoliated urinary cells could detect urinary tract malignancy.
- **Sensitivity 31–62%, specificity 94–100%**; reporting varies between institutions, compounding a subjective test.
- **The Paris System for Reporting Urinary Cytology** (introduced 2013 at the International Congress of Cytology) standardises categories and cytomorphological criteria for **high-grade urothelial carcinoma (HGUC)**; it appears to **reduce inconclusive "atypia" calls** and improve risk stratification.
- **Positive or suspicious high-grade cytology must be investigated** because the test is highly specific.
- **Negative cystoscopy with persistently positive high-grade cytology → random biopsies of bladder and prostatic urethra**, plus **ureteral washings for cytology**, with or without ureteroscopic inspection of the upper tracts.

Fig. 129.6 — *The Paris System for Reporting Urinary Cytology, version 2.0* (Barkan et al., 2016). Algorithm:

- **Nuclear-to-cytoplasmic (N:C) ratio ≥ 0.5 ?** (excluding intermediate and basal cells) → No: **NHGUC**. Yes → assess **hyperchromasia, coarse chromatin, irregular chromatin rim** → atypia explained by non-neoplastic change (polyomavirus, granuloma, ileal conduit, urolithiasis, reactive change): NHGUC → **≥ 1 feature: AUC** → **≥ 2 features and N:C ≥ 0.7 : few cells ($<5-10$) = suspicious for HGUC; many = HGUC.**

PARIS SYSTEM CATEGORIES

Category	Criteria
Negative for HGUC (NHGUC)	N:C <0.5 , or atypia commensurate with a non-neoplastic cause
Atypical urothelial cells (AUC)	N:C ≥ 0.5 and at least one feature
Suspicious for HGUC	N:C ≥ 0.7 , at least two features, few cells ($<5-10$)
HGUC	N:C ≥ 0.7 , at least two features, many cells

Urine-based biomarkers

- **Adjuncts, not replacements:** developed because cystoscopy is costly and uncomfortable; most are **more sensitive than cytology but less specific**.
- **Nuclear matrix protein 22 (NMP22).** A nuclear structural protein about **20 times more abundant in malignant urothelial cells**. Two FDA-approved tests for surveillance: a **laboratory quantitative** assay and a **point-of-care qualitative** assay; accuracy is similar at diagnosis and in surveillance.

Fluorescence in Situ Hybridization

- **UroVysion** fluorescence in situ hybridisation (FISH) uses labelled DNA probes to detect **aneuploidy of chromosomes 3, 7 and 17** and **homozygous loss of 9p21** in exfoliated cells.

- **Positive test:** ≥ 5 cells with gains of ≥ 2 chromosomes, or ≥ 10 cells with gain of a single chromosome, or homozygous 9p21 deletion in $>20\%$ of cells.
- **Reflex test after atypical cytology:** a negative FISH correlates with benign cytological change; with atypical cytology and negative cystoscopy, a **positive FISH predicts lower recurrence-free survival**.
- **Limitations:** low sensitivity for **low-grade tumours**; no consensus on criteria for abnormal cells.

Bladder Tumor Antigen

- **Bladder tumour antigen (BTA)** assays detect **complement factor H-related protein and complement factor H** with monoclonal antibodies; both forms are FDA approved for diagnosis and follow-up.
- **BTA stat:** qualitative, dipstick, point of care. **BTA TRAK:** quantitative, enzyme-linked immunosorbent assay (ELISA).
- **High false-positive rate with haematuria:** complement factor H is abundant in serum, so blood cross-reacts.

ImmunoCyt

- **A cell-based adjunct to cytology** (late 1990s): fluorescent antibodies against a **glycosylated carcinoembryonic antigen (CEA) and two mucins**.
- **Combined with cytology it raises sensitivity but lowers specificity.**
- **Operator dependent**, with high interobserver variability; adequate training may overcome this.

CxBladder

- **A cell-based assay of five messenger RNA fragments:** HOXA13, CDC2, MDK, CXCR2, IGFBP5.
- **Sensitivity and specificity about 85%** in the body text versus 0.82 and 0.85 in Table 129.1 — the table figure is a single study at a threshold fixed for specificity 0.85, the text figure a summary of the few published studies. Early comparisons suggest it outperforms cytology and protein assays; it **needs independent multi-institutional validation**.

TABLE 129.1 TEST PERFORMANCE OF URINARY BIOMARKERS FOR DIAGNOSIS OF BLADDER CANCER (*Data from Chou et al., 2015 — systematic review and meta-analysis*)

Assay	Target / format	Overall sens / spec	Symptom evaluation sens / spec	Surveillance sens / spec
NMP22, quantitative	Nuclear matrix protein; laboratory	0.69 / 0.77	0.67 / 0.84	0.61 / 0.71
NMP22, qualitative	Point of care	0.58 / 0.88	0.47 / 0.93	0.70 / 0.83
BTA, qualitative (stat)	Complement factor H $\pm$ related protein; dipstick	0.64 / 0.77	0.76 / 0.78	0.60 / 0.76
BTA, quantitative (TRAK)	ELISA	0.65 / 0.74	0.76 / 0.53	0.58 / 0.79
FISH (UroVysion)	Chromosomes 3, 7, 17; 9p21	0.63 / 0.87	0.73 / 0.95	0.55 / 0.80
ImmunoCyt	CEA + two mucins; cell based	0.78 / 0.78	0.85 / 0.83	0.75 / 0.76
CxBladder	Five-mRNA panel	—	0.82 / 0.85*	—

- **What the table is for:** to show that no biomarker approaches cytology's specificity while beating its sensitivity by a margin that would displace cystoscopy. *CxBladder threshold was chosen for a specificity of 0.85.

- **Pattern:** ImmunoCyt has the highest sensitivity; FISH and qualitative NMP22 the highest specificity; performance is generally **better when evaluating symptoms than in surveillance**, where tumours are smaller and lower grade. FISH in symptomatic patients gives the highest positive likelihood ratio.
- **The trap:** a negative marker does not exclude cancer. Head-to-head comparisons are too limited to rank the assays.
- **Authors' practice:** in microhaematuria, adding a biomarker **does not remove the need for cystoscopy**; in gross haematuria a negative result still leaves a **≥10% probability of cancer**, so cystoscopy is still required. Biomarkers are used **with cystoscopy in NMIBC surveillance**; multiplex gene assays await validation.

On the Horizon: Urinary Tumor DNA

- **Urinary tumour DNA (utDNA)** enables urinary genomic sequencing. High levels track **tumour burden and stage** and predict **minimal residual disease and local recurrence**.
- **Its appeal:** it reflects the **molecular profile of the primary tumour**, so it could tailor therapy and detect early recurrence.

KEY POINTS

- **Painless gross haematuria** is the key presenting symptom, in **85%** of new diagnoses.
- A full haematuria evaluation comprises **focused history and examination, cystoscopy, upper tract imaging and urine culture**.
- The **gold standard is cystoscopy and biopsy**; HAL-BLC and NBI are increasingly used as adjuncts for occult malignancy.
- **No urine marker** yet has high enough sensitivity or specificity to replace office cystoscopy.

Concept — Cystoscopy remains indispensable because every adjunct trades one property for another: cytology is specific for high-grade disease but blind to low grade, and urine markers are more sensitive but not specific enough — so a **positive adjunct triggers investigation and a negative one never excludes**.

PATHOLOGY

- **Numerous benign bladder tumours** exist, each with its own presentation and disease burden.

Epithelial Metaplasia

- A **focal area of transformed urothelium with normal nuclear and cellular morphology**, typically on the **trigone**; squamous or glandular.
- **Glandular metaplasia:** raised, red, inflammatory clumps that can be **mistaken for cancer**.
- **Squamous metaplasia:** common — up to **40% of females and 5% of males**; associated with UTI, trauma and prior surgery.
- **No biopsy needed; no treatment required or effective.**

Papilloma and Inverted Papilloma

- **Inverted papilloma:** benign proliferative lesion associated with **chronic inflammation or bladder outlet obstruction**; anywhere in the bladder but **commonest on the trigone**; **<1% of bladder tumours**. Presents with painless gross haematuria (commonest), microscopic haematuria or irritative symptoms.
- **Endophytic growth** into the lamina propria forming nests and anastomosing trabeculae. When strictly defined (**no cytological atypia**) it behaves benignly, with about **1% recurrence**.

- **Urothelial papilloma:** delicate stalks lined by normal-appearing urothelium; **formerly classed as low-grade Ta cancer, now benign** with no invasive risk. It carries **fibroblast growth factor receptor 3 (FGFR3) mutations but lacks TP53 and RB mutations**.

Nephrogenic Adenoma

- **Caused by chronic urothelial irritation:** trauma, prior surgery, **renal transplantation**, intravesical chemotherapy, stones, catheters, infection.
- **Glandular tubules resembling renal tubules** in the mucosa and submucosa; cystoscopically may look papillary and **mimic an aggressive neoplasm**.
- Causes dysuria and haematuria; **recurrence is uncommon and managed by repeat resection**.

Leukoplakia

- **Keratinising squamous metaplasia** with **white, flaky plaques floating in the bladder**; if not incidental, presents with recurrent infection, urgency and frequency. The mucosa is chronically inflamed.
- **Premalignant on other squamous surfaces, but not in the bladder:** historical reports called it premalignant, but DNA analysis for TP53 found no evidence of premalignancy, and it is no longer an independent risk factor → **further cystoscopy, biopsy and treatment are unnecessary**.

Cystitis Cystica and Glandularis

- **Common in normal bladders**, associated with inflammation or chronic obstruction; cystic nests lined by columnar or cuboidal cells, arising from proliferating **von Brunn nests**.
- **Cystitis glandularis can transform to adenocarcinoma**, though the risk is low → **regular endoscopic evaluation**.
- **Treatment:** transurethral resection and relief of the obstruction or inflammation.

BENIGN BLADDER LESIONS COMPARED

Lesion	Setting	Hallmark	Malignant potential	Management
Epithelial metaplasia	Trigone; UTI, trauma, surgery	Normal nuclei; glandular type mimics cancer	None	Nothing
Inverted papilloma	Trigone; inflammation, outflow obstruction	Endophytic nests, no atypia	Benign; 1% recurrence	Resection
Urothelial papilloma	—	Delicate stalks, normal urothelium; FGFR3+, TP53/RB wild type	None (formerly LG Ta)	Resection
Nephrogenic adenoma	Irritation, transplant, catheters	Renal tubule-like glands	None; recurrence uncommon	Repeat resection
Leukoplakia	Chronic inflammation	Floating white keratin flakes	Not premalignant (disproven)	No further treatment
Cystitis glandularis	Inflammation, obstruction	Von Brunn nest proliferation	Low — adenocarcinoma	Resect, treat cause, endoscopic surveillance

Concept — *Benign bladder lesions are almost all **reactive responses to chronic irritation**; the clinical task is to recognise the mimic, and only cystitis glandularis retains a malignant potential worth surveillance.*

PRECURSOR MALIGNANT LESIONS

- **Normal urothelium:** four to seven cells thick, transitional between nonkeratinising squamous and pseudostratified columnar epithelium, maturing in order from **small cuboidal basal cells** → **intermediate cells** → **superficial umbrella cells**. Umbrella cells may show nuclear atypia and form the **urine–blood barrier** that protects deeper cells from toxins.
- **A continuum:** hyperplasia → atypia → dysplasia → cancer. The **2004 WHO classification** lists premalignant lesions as urothelial hyperplasia (flat and papillary), reactive atypia, urothelial atypia of unknown significance, urothelial dysplasia and low-grade intraurothelial neoplasia.
- **Urothelial hyperplasia:** flat or papillary; mucosa **≥10 cell layers**; nuclei may be slightly enlarged but **no atypia and no mitoses**. Not malignant in isolation but often accompanies neoplasia. Flat hyperplasia may precede **low-grade papillary tumours** (FGFR3 mutation; loss of heterozygosity at 3q, 5q, 10q, 13q and 17p). **Papillary hyperplasia is cystoscopically indistinguishable** from, and often adjacent to, low-grade papillary tumours.
- **Reactive atypia:** nuclear abnormality on an **inflamed background** (chronic inflammation, regeneration, noxious stimuli), presenting with irritative symptoms; oedematous and inflamed endoscopically. Nuclei may be enlarged, but **maturation and nuclear polarity are preserved**. Not independently malignant, but **coexists with dysplasia and CIS** → **investigate rigorously**.
- **Atypia of unknown significance (AUS):** used when reactive versus premalignant is unclear; lamina propria inflammation with **atypia disproportionate to the inflammation**. **CK20 and CD44 immunostaining** help separate it from dysplasia. Not independently progressive; **limited clinical utility**.
- **Urothelial dysplasia:** **normal thickness** with loss of cytoplasmic clearing (basal and intermediate cells), **loss of polarity**, nuclear enlargement, hyperchromasia and membrane irregularity; lamina propria usually normal. **CK20 and p53 staining** highlight dysplastic cells. **Umbrella cells are retained**. A marker of urothelial instability: **14–19% progress de novo**, but with a prior bladder cancer **dysplasia leads to CIS in 60%**.

FLAT UROTHELIAL LESIONS COMPARED

	Hyperplasia	Reactive atypia	AUS	Dysplasia	CIS
Thickness	≥10 layers	Variable, oedematous	Variable	Normal	Variable
Atypia	None	Proportionate to inflammation	Disproportionate to inflammation	Moderate	Severe
Polarity / maturation	Preserved	Preserved	—	Lost	Lost; noncohesive
Umbrella cells	Present	Present	Present	Present	Lost
Staining	—	—	CK20, CD44	CK20, p53	—
Malignant potential	None alone	None alone	None	14–19%; 60% → CIS if prior cancer	Precursor of high-grade invasion

Concept — Flat lesions are graded on two questions: **is the atypia out of proportion to inflammation, and are maturation, polarity and umbrella cells preserved?** Loss of polarity marks dysplasia; loss of umbrella cells marks CIS.

MALIGNANT LESIONS

- **Urothelial carcinoma is the commonest malignancy of the urinary tract and the second commonest cause of death among genitourinary tumours.**
- **NMIBC comprises about 70% of bladder urothelial carcinoma:** low- and high-grade noninvasive papillary tumours, CIS, and lamina propria invasion (T1).
- **CIS:** flat, noninvasive, **high grade**; severe atypia, loss of polarity, noncohesive large pleomorphic cells with clumped chromatin and abnormal mitoses. **Loss of umbrella cells separates it from dysplasia.** Its genetics (**RB, TP53, PTEN**) resemble high-grade invasive tumours — hence a **precursor of invasive carcinoma** — and it has **no FGFR3 mutations**, a lineage distinct from low-grade papillary tumours. CIS with an invasive tumour carries a **45–65% 5-year mortality**.
- **Papillary urothelial neoplasm of low malignant potential (PUNLMP):** very good prognosis, and whether it is "cancer" is debated. Papillary, >7 cells thick, minimal atypia, generally solitary, typically on the trigone; polarity maintained, nuclei mildly enlarged. Recurrences are nearly always PUNLMP or low grade; rare high-grade recurrences may be new tumours rather than progression.
- **Low-grade urothelial carcinoma:** papillary with a fibrovascular stalk and frequent branching, enlarged cells, some atypia, occasional mitoses; **chromosome 9 alterations almost universal**, with FGFR3, PI3K or RAS mutations.
- **High-grade papillary carcinoma:** fused papillae, disordered growth, numerous mitoses, pleomorphic cells with exaggerated nuclei; **more than 80% invade the stroma if untreated** — a precursor of invasive high-grade cancer.

THE NONINVASIVE SPECTRUM COMPARED

	PUNLMP	Low-grade Ta	High-grade papillary	CIS
Architecture	Papillary, >7 cells	Papillary, branching stalks	Fused papillae, disordered	Flat
Atypia	Minimal	Some	Marked	Severe
Genetics	TERT promoter and FGFR3 in >60%; HRAS low (3% vs 33%)	Chromosome 9; FGFR3, PI3K or RAS	—	RB, TP53, PTEN; no FGFR3
Recurrence	About 20% at 5 years	48–71%	—	—
Progression	<1% to MIBC	2–12%	>80% invade if untreated	45–65% 5-year mortality with invasive tumour

Concept — *Two lineages run through the noninvasive spectrum: **FGFR3-driven papillary tumours** (PUNLMP, low-grade Ta) recur but rarely progress, whereas **TP53/RB-driven high-grade tumours** (CIS, high-grade papillary) progress. Grade, not recurrence, predicts danger.*

STAGING

- **American Joint Committee on Cancer (AJCC)**, with the Union for International Cancer Control (UICC), sets the tumour, nodes, metastases (TNM) classification; the **2017 system** is current.
- **Stage groups in outline:** NMIBC = stage 0 (noninvasive) and stage I (subepithelial connective tissue); **stage II** muscle-invasive, organ confined; **stage III** muscle-invasive, locally advanced; **stage IV** metastatic.
- **2017 change:** nodal disease without systemic metastasis is now **stage III**; earlier editions made any positive node stage IV.

- **T1 subcategorisation into T1a (superficial) and T1b (deep) lamina propria invasion**, relative to the muscularis mucosae, is recommended to stratify a heterogeneous group: about **50% of T1 tumours are upstaged to ≥T2** and **about a third to non-organ-confined disease** at cystectomy. **Deeper lamina propria invasion means worse survival.**
- **T2a (inner half) versus T2b (outer half) of the muscularis propria**: prognostic significance is controversial and complicated by TURBT sampling, so both are treated alike (stage II).
- **T3**: perivesical fat — **T3a microscopic, T3b macroscopic. T4a**: prostatic stroma, uterus or vagina. **T4b**: pelvic or abdominal wall.

NMIBC RISK GROUPS

Risk group	Definition
Low	Primary (not recurrent), solitary, low-grade Ta, <3 cm
Intermediate	Low-grade Ta that is multiple and/or recurrent and/or large (>3 cm)
High	Any high-grade feature — CIS or T1

TABLE 129.2 UPDATED 2017 AMERICAN JOINT COMMITTEE ON CANCER STAGING SYSTEM (Amin *et al.*, 2017)

Stage	T	N	M	Meaning
0a	Ta	N0	M0	Noninvasive papillary
0is	Tis	N0	M0	Flat CIS, confined to the lining
I	T1	N0	M0	Subepithelial connective tissue, not muscle
II	T2a or T2b	N0	M0	Inner or outer muscle, not through it
IIIA	T3a, T3b, T4a	N0	M0	Perivesical fat, or prostate, uterus, vagina
	T1–T4a	N1	M0	Single node in the true pelvis
IIIB	T1–T4a	N2 or N3	M0	≥2 true pelvic nodes (N2) or common iliac nodes (N3)
IVA	T4b	Any N	M0	Pelvic or abdominal wall
	Any T	Any N	M1a	Distant lymph nodes
IVB	Any T	Any N	M1b	Distant organs — bone, liver, lung

- **What the table is for**: to fix where each anatomical boundary falls. **Pattern**: T1–T4a with any regional node is stage III; only T4b, distant nodes (M1a) or organ metastases (M1b) are stage IV. **Trap**: common iliac nodes are N3 (stage IIIB), not M1a.

Prostatic urethral involvement

- **Prostatic urethral urothelial cancer accompanies bladder cancer in 90%** of cases, but among primary bladder cancers **urethral involvement is rare (about 3%)**; most is CIS.
- **Mucosal extension without stromal invasion is not adverse**, but it is hard to treat with intravesical therapy — hence more prostatic urethral CIS in patients receiving frequent intravesical therapy.
- **Prostatic stromal invasion**, especially direct transmural extension from the bladder, carries a poor prognosis (historic 5-year overall survival <25%) → **only stromal invasion is T4**.

Concept — Local stage turns on two muscle landmarks — the **muscularis mucosae** (T1a versus T1b) and the **muscularis propria** (NMIBC versus MIBC) — and since 2017 regional nodes are a **regional (stage III)**, not systemic (stage IV), **problem**.

MOLECULAR BIOLOGY

- **Recurrent alterations:** FGFR3, the phosphoinositide 3-kinase (PI3K) pathway, mitogen-activated protein kinase (MAPK) signalling, cell cycle regulation, and hedgehog and WNT signalling.
- **Cell cycle genes**, mutated especially in MIBC: **RB1 and CDKN2A inactivation** is common; **p53 inactivation in up to 76% of MIBC** (The Cancer Genome Atlas, TCGA); RB1 loss drives amplification and overexpression of the oncogene **E2F3**.
- **FGFR3 — the NMIBC hallmark**, with deletions of 9p and 9q: activating point mutations in **up to 80% of low-grade Ta** versus **10–20% of MIBC**. Normal urothelium expresses the **IIIb isoform** (binds FGF1); a **switch to IIIc** may drive oncogenesis by autocrine and paracrine signalling. **FGFR3 fusions with TACC3 or BAI1** are highly activated oncoproteins.
- **MAPK — RAS** (HRAS or KRAS): first identified in the **T24 bladder cancer cell line**; **mutually exclusive with FGFR3** because both lie on the RAS–MEK–ERK pathway; unlike FGFR3, **equally frequent in NMIBC and MIBC**.
- **PI3K — PIK3CA** point mutations in **25% of NMIBC**, fewer in MIBC; they **co-occur with FGFR3**. PTEN negatively regulates PI3K; **PTEN loss in MIBC accompanies TP53 alteration and poor prognosis**. Dual deletion of p53 and PTEN produces metastatic bladder cancer in transgenic mice; PTEN's **protein** (not lipid) phosphatase activity restrains invasion and metastasis.
- **WNT/β-catenin** cooperates with PI3K/PTEN: organoid cultures show its role in proliferation, and activated β-catenin with PTEN deletion promotes bladder cancer.

DRIVER GENES — NMIBC VERSUS MIBC

Gene	Pathway	NMIBC	MIBC	Key point
FGFR3	Receptor tyrosine kinase → RAS–MEK–ERK	Up to 80% (low-grade Ta)	10–20%	IIIb → IIIc switch; TACC3/BAI1 fusions
HRAS / KRAS	MAPK	Equal		Mutually exclusive with FGFR3
PIK3CA	PI3K	25%	Less	Co-occurs with FGFR3
TP53	Cell cycle	—	Up to 76%	With PTEN loss → metastasis
RB1, CDKN2A	Cell cycle	—	Common	RB1 loss → E2F3 overexpression
PTEN	PI3K (negative regulator)	—	Downregulated	Poor prognosis

THE CLASSICAL TWO-PATHWAY MODEL

	Papillary pathway	Flat pathway
Origin	Urothelial proliferation (hyperplasia)	Dysplasia
Product	Low-grade noninvasive papillary	CIS → high-grade invasive
Genetics	FGFR3 ; low total mutational burden	TP53, RB

- **Superseded by expression subtypes:** genome-wide expression and sequencing have identified clear driver genes and molecular subclasses that are **agnostic to conventional stage and grade**.
- **TCGA (2014):** mRNA, microRNA and protein profiling of MIBC identified **four clusters resembling breast cancer intrinsic subtypes**, confirmed by **Lund, MD Anderson (MDA) and the University of North Carolina (UNC)**; the systems overlap along a **luminal–basal axis**.
- **MDA "p53-like":** a luminal subgroup that is **chemoresistant** with poor prognosis. **UNC "claudin-low":** enriched for RB1, EP300 and NCOR1, with low FGFR3, KDM6A and PPARG; **immune infiltrated yet actively immunosuppressed** → candidates for FDA-approved **checkpoint inhibitors**.

- **TCGA follow-up (412 MIBC): five expression subtypes** (Fig. 129.15) and an **APOBEC-driven high-mutational-load subset** (APOBEC: cytidine deaminases involved in mRNA editing) with markedly **better 5-year survival** than typical MIBC.
- **Consensus classification (2020):** reconciled existing schemas into **six classes** (Fig. 129.16).

LUMINAL VERSUS BASAL PHENOTYPE

	Luminal	Basal / squamous
Markers	High FGFR3, GATA3, KRT20, PPARG expression	CK5, CK6, CK14; low FOXA1 and GATA3
Transcription factors	PPARG, RXRA, GATA3, FOXA1	STAT3, FOXM1
Actionable targets	FGFR3, ERBB2	EGFR; immune checkpoints
Behaviour	Less aggressive	More aggressive; squamous differentiation

Fig. 129.14 — Molecular subtypes compared across classification systems (Comp  rat et al., 2018).

- **UNC** (luminal, basal), **MDA** (luminal, p53-like, basal), **TCGA 2014** (clusters I–IV) and **Lund** (urothelial-like A, genomically unstable, infiltrated, SCC-like, urothelial-like B) map onto one another; **Lund alone covers NMIBC as well as MIBC**.

FIG. 129.15 TCGA EXPRESSION SUBTYPES AND PROPOSED THERAPY (Robertson et al., 2017)

Subtype	Features	Proposed approach
Luminal-papillary	FGFR3 mutation, fusion, amplification; papillary histology; sonic hedgehog (SHH)+; low CIS; low risk	FGFR3 inhibitors ; neoadjuvant chemotherapy (NAC) — low predicted response
Luminal-infiltrated	Low purity; epithelial–mesenchymal transition (EMT) markers; miR-200; medium PD-L1, CTLA-4; myofibroblast markers; wild-type p53	Anti-PD-L1, PD-1, CTLA-4 ; cisplatin NAC has a low response rate
Luminal	Uroplakins, KRT20, SNX31	Targeted therapy (uncertain)
Basal/squamous	KRT5/6/14+, GATA3–, FOXA1–; female ; squamous differentiation; high PD-L1, CTLA-4; immune infiltrates	Cisplatin-based NAC ; anti-PD-L1, PD-1, CTLA-4
Neuronal	SOX2, DLX6, MSI1, PLEKHG4B; E2F3/SOX4 amplification; high cell cycle	Etoposide–cisplatin NAC

FIG. 129.16 CONSENSUS CLASSIFICATION OF MOLECULAR SUBTYPES OF BLADDER CANCER (Kamoun et al., 2020)

Class (% of MIBC)	Differentiation	Key alterations	Infiltrate / histology / clinical	Median OS (years)
Luminal papillary (24%)	Urothelial/luminal	FGFR3+, PPARG+, CDKN2A–; FGFR3 40%, KDM6A 38%	Papillary morphology; T2 stage	4
Luminal nonspecified (8%)	Urothelial/luminal	PPARG+; ELF3	Fibroblasts; micropapillary variant ; older patients (80+)	1.8
Luminal unstable (15%)	Urothelial/luminal	PPARG+, E2F3+, ERBB2+; genomic instability; TP53, ERCC2; high tumour mutational burden, APOBEC	—	2.9
Stroma-rich (15%)	Intermediate	—	Smooth muscle, fibroblasts, myofibroblasts; B cells	3.8
Basal/squamous (35%) — commonest	Basal	EGFR+; TP53, RB1	CD8 T cells, natural killer cells; squamous differentiation; women, T3/T4	1.2

Class (% of MIBC)	Differentiation	Key alterations	Infiltrate / histology / clinical	Median OS (years)
Neuroendocrine-like (3%)	Neuroendocrine	TP53 and RB1 loss (94% have RB1 mutation or deletion); cell cycle+	Neuroendocrine differentiation	1 — worst

- **What the table is for:** to link subtype to behaviour and treatment. **Pattern:** survival falls from luminal papillary (best) to basal/squamous and neuroendocrine-like (worst). **Trap:** the micropapillary variant sits in a luminal class, and basal/squamous — the commonest class — is enriched in women.

KEY POINTS

- Genome-wide expression and sequencing have identified **clear driver genes and pathways**.
- An emerging **molecular taxonomy is broadly defined by urothelial cell type**.
- **Luminal:** high levels of FGFR3, GATA3, KRT20 and PPARG.
- **Basal:** squamous differentiation, CK5, CK6 and CK14 expression, and low FOXA1 and GATA3 RNA and protein.

Concept — *The papillary-versus-flat model has given way to **expression subtypes along a luminal–basal axis** that predict behaviour and candidate therapy independent of stage and grade: luminal papillary is indolent and FGFR3-targetable; basal/squamous is aggressive but immune-infiltrated and cisplatin-responsive; neuroendocrine-like is worst.*

HISTOLOGIC VARIANTS OF UROTHELIAL CARCINOMA

- **Formerly "transitional cell carcinoma"** because of its propensity to differentiate into other tumour types — SCC, adenocarcinoma, clear cell carcinoma.
- **The variant spectrum** now includes distinct growth patterns, altered differentiation, stromal reactions and unique responses to treatment and external stimuli.

Micropapillary Variant

- **0.7–2.2% of urothelial tumours** (first described 1994); **strong male predominance**.
- **Advanced at diagnosis:** more than half are T3 or T4, and survival is lower than with other histologies. Even at lower stage it carries a **high progression rate from NMIBC to MIBC** and high subsequent metastasis despite treatment.
- **Histology:** small tight clusters or thin papillae of atypical cells with eosinophilic to clear cytoplasm, surrounded by **prominent retraction artefact**.
- **Any amount of micropapillary component defines the tumour** as micropapillary and dictates management.
- **Surgery is the most effective therapy at all stages:** transurethral resection plus bacillus Calmette–Guérin (BCG) is ineffective unless the tumour is completely resected, and **early cystectomy is encouraged for NMIBC** (better cancer-specific survival than TUR plus BCG). In MIBC, **NAC appears ineffective → immediate cystectomy**.
- **Molecular:** **ERBB2 amplification**, activation of miR-296 and RUVBL1 targets; evolves through a **luminal pathway** (PPAR-γ enrichment, p63 suppression); a subset with **wild-type p53 activation** is the most aggressive.

Sarcomatoid Variant

- **0.3–0.6% of bladder cancers**; carcinosarcoma and sarcomatoid carcinoma, once separate, share a common clonal origin.
- **Large infiltrative masses** at higher stage; poor cancer-specific survival, and **worse than pure urothelial carcinoma stage for stage**.
- **Poor chemotherapy response → immediate radical cystectomy** where feasible, sometimes with intraoperative radiation therapy (IORT).

Plasmacytoid Variant

- **<1% of urothelial tumours**; the **2016 WHO classification includes the signet ring variant** (cytoplasmic vacuoles with or without mucin).
- **Histology**: eosinophilic cytoplasm with eccentric, enlarged, hyperchromatic nuclei.
- **Advanced at presentation**, with deep muscle and perivesical invasion and characteristic **intraperitoneal spread**; driven by **truncating CDH1 alterations** (84% in one study).
- **Chemosensitive → NAC whenever possible**, but relapse is common; median survival in nonmetastatic disease about 1.5 years (17.7 months).

Nested Variant

- **Rare but aggressive**; male-to-female ratio **6:1**.
- **Mimics benign lesions**: von Brunn nests, cystitis cystica, inverted papilloma.
- **Typically upstaged at cystectomy and locally advanced at presentation**, yet **stage-matched outcomes resemble pure urothelial carcinoma**.

Urothelial Carcinoma With Divergent Differentiation

- **Squamous differentiation** is reported in 16–22% and probably underestimated — up to 60% have squamous elements; underreporting obscures its prognostic effect. It is associated with the **basal subtype**, more advanced stage and focal aggressiveness, and with recurrence and progression.
- **Glandular differentiation** (small tubular or gland-like spaces) in about **10%**: associated with high stage and clinical-to-pathological upstaging, but **recurrence-free and overall survival match conventional urothelial carcinoma**.

HISTOLOGICAL VARIANTS COMPARED

Variant	Frequency	Hallmark	Molecular	Chemosensitive?	Management
Micropapillary	0.7–2.2%	Retraction artefact; any amount counts	ERBB2 amplification; luminal	No	Early cystectomy, even NMIBC
Sarcomatoid	0.3–0.6%	Large infiltrative mass	Common clonal origin	Poor	Immediate cystectomy ± IORT
Plasmacytoid	<1%	Eccentric nuclei; intraperitoneal spread ; includes signet ring	CDH1 truncation	Yes (relapse common)	NAC whenever possible
Nested	Rare; M:F 6:1	Mimics von Brunn nests	—	—	As stage-matched urothelial carcinoma
Squamous differentiation	16–22% (up to 60%)	Basal subtype	Basal	—	Associated with recurrence and progression

Variant	Frequency	Hallmark	Molecular	Chemosensitive?	Management
Glandular differentiation	About 10%	Gland-like spaces	—	—	Outcome as conventional

Concept — Altered growth patterns (*micropapillary, sarcomatoid, plasmacytoid*) behave worse than their stage and push towards early surgery or variant-specific chemotherapy, whereas **divergent differentiation and the nested pattern** mostly reflect stage at presentation. Chemosensitivity is variant specific: *plasmacytoid responds; micropapillary and sarcomatoid do not.*

NONUROTHELIAL MALIGNANCY

Small Cell Carcinoma

- **A rare, poorly differentiated neuroendocrine neoplasm**, primarily of lung but also prostate and bladder.
- **Presentation**: usually haematuria; paraneoplastic **hypercalcaemia, Cushing syndrome and sensory neuropathy** reported.
- **Histology (Fig. 129.19)**: sheets and nests of basophilic cells with high nuclear-to-cytoplasmic ratio, resembling lung small cell; **chromogranin and synaptophysin positive**.
- **Genetics**: high somatic mutational burden (APOBEC-mediated); frequent **TP53, RB1 and TERT promoter** mutations. Evolutionary studies suggest **dedifferentiation from urothelial carcinoma**, not a shared lineage with lung small cell.
- **Aggressive and chemosensitive → platinum-based chemotherapy first, regardless of stage; cisplatin plus etoposide** (the lung regimen) is the treatment of choice.
- **Sequence matters**: primary cystectomy gives poor disease-free survival not rescued by adjuvant chemotherapy, whereas **neoadjuvant chemotherapy yields markedly better cancer-specific survival**. One series found stage-matched outcomes similar to urothelial carcinoma, with a possible adjuvant role.
- **Chemoradiation**, historically preferred and still advocated by some as trimodal therapy, is less supported: **chemotherapy plus cystectomy gives the highest response rates**, and chemoradiation needs further study.

Squamous Cell Cancer

- **Chronic infection** with *S. haematobium* or other bacteria causes SCC; in **endemic countries SCC is the majority** of bladder cancer, with its own growth pattern and algorithm. In the US and Europe it is 2–5%.
- **Chronic inflammation is central**: SCC in catheterised spinal cord injury patients has fallen substantially with the move to **intermittent self-catheterisation**, supporting the role of infection and foreign bodies.
- **Presentation**: **fewer nodal and distant metastases but more advanced local stage** than urothelial carcinoma.
- **Radical cystectomy is the mainstay**; NAC has no consensus. **Preoperative radiation worsened survival** (1970s–1980s) → use only if cystectomy is not planned. Adjuvant therapy and IORT lack consensus.
- **Molecular data** are limited, mainly schistosomal; **FGF2** is associated with aggressive features and worse outcome after cystectomy.

Adenocarcinoma

- **Primary** (pure glandular phenotype from bladder urothelium) or **secondary** — metastasis or, **more commonly, direct extension from colon, prostate, endometrium or cervix**. Urachal adenocarcinoma shares pathological and clinical features.

- **Risk factors:** bladder exstrophy, schistosomiasis, chronic irritation or obstruction.
- **Secondary is commoner than primary → exclude another source;** the authors perform **colonoscopy whenever adenocarcinoma is found**. Histology shows cribriform and glandular architecture (Fig. 129.21).
- **Most primary tumours are muscle invasive → radical cystectomy** is standard. There are **no level 1 data** for neoadjuvant or adjuvant chemotherapy or radiation; methotrexate, vinblastine, doxorubicin (Adriamycin) and cisplatin (MVAC) or gemcitabine–cisplatin (GC) give little benefit; **5-fluorouracil plus cisplatin** is often used by analogy with other adenocarcinomas.

Urachal Adenocarcinoma

- **About one-third of bladder adenocarcinomas;** arises from the **allantoic remnant** connecting bladder to umbilical cord. Although urachal remnants are lined by urothelium, the cancer is **almost always adenocarcinoma**.
- **2016 WHO diagnostic criteria:** located in the **dome or anterior wall** with the **epicentre in the bladder wall**; **no widespread cystitis cystica/glandularis**; **secondary source excluded**.
- **TNM is unsuitable** (nearly all are muscle invasive by location) → **Sheldon staging**.
- **Treatment:** **en bloc resection of the bladder dome, urachal ligament and umbilicus**.

SHELDON STAGING OF URACHAL CARCINOMA

Stage	Extent
I	Confined to urachal mucosa
II	Confined to the urachus
IIIA / IIIB / IIIC	Local extension into bladder / abdominal wall / peritoneum
IV	Metastatic

NONUROTHELIAL MALIGNANCIES COMPARED

	Small cell	Squamous cell	Adenocarcinoma	Urachal adenocarcinoma
Frequency / setting	Rare	2–5% West; majority where schistosomiasis endemic	Secondary > primary	One-third of adenocarcinomas
Aetiology	Dedifferentiated urothelial carcinoma	Chronic infection, catheters	Exstrophy, schistosomiasis, irritation	Allantoic remnant
Hallmark	Chromogranin, synaptophysin; paraneoplastic	Locally advanced, few metastases	Must exclude colon, prostate, uterine source	Dome; Sheldon staging
First-line treatment	Cisplatin–etoposide first , then cystectomy or radiation	Radical cystectomy	Radical cystectomy	En bloc dome, urachus, umbilicus
Chemotherapy / radiation	Chemosensitive	No consensus; preoperative radiation harmful	No level 1 data; 5-fluorouracil–cisplatin	—

KEY POINTS

- **Eighty percent** of urothelial carcinomas contain some mixed differentiation, most commonly **squamous**.

- **Altered growth patterns**, particularly micropapillary, carry a poor prognosis even in non-muscle-invasive disease; the nested variant presents at advanced stage.
- **NAC** appears active in mixed tumours containing adenocarcinoma or SCC, but **not against altered growth patterns** such as micropapillary and nested urothelial cancer.
- **Small cell carcinoma should be treated as metastatic disease**: chemotherapy first, then radiation or surgery for local control.

Concept — *Each nonurothelial cancer is defined by its origin: **small cell is treated systemically first**, SCC and adenocarcinoma **surgically first**, and any bladder adenocarcinoma must be **proven primary before it is treated as bladder cancer**.*